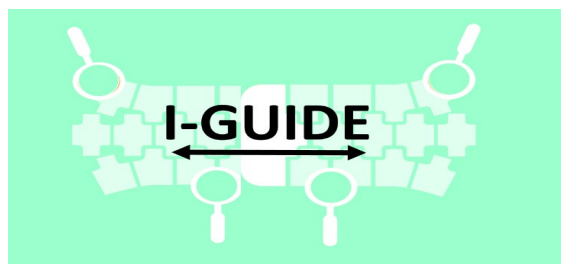


Title: I-GUIDE: Image guided minimally invasive interventions


1. General information

1.a Applicants

In case an applicant has more than one affiliation, please mention this in the table.

<i>Name*</i>	<i>Institution</i>	<i>Department</i>	<i>Email</i>	<i>Tenured (i.e. fixed) position/ Tenure-track/</i>	<i>For Erasmus MC applicants: clinician/ researcher/ mix</i>	<i>Role in Flagship [Main Lead/Lead / Applicant]</i>
Prof. dr. Aad (A.) van der Lugt	Erasmus MC	Radiology & Nuclear Medicine	a.vanderlugt@erasmusmc.nl	Tenured	Clinician/ Researcher	Main Lead
Prof. dr. Jenny (J.) Dankelman	TU Delft	Biomechanical Eng.	j.dankelman@tudelft.nl	Tenured		Lead
Dr. Sandra Sülz	Erasmus University ESHPM	Health Service Management and Organisation	sulz@eshpm.eur.nl	Tenured		Lead
Prof. dr. Gijs (G.) van Soest	Erasmus MC	Cardiology	g.vansoest@erasmusmc.nl	Tenured	Researcher	Applicant
Prof. dr. ir. Ton (A.F.W.) van der Steen	Erasmu sMC	Cardiology/ Biomedical Engineering	a.vandersteen@erasmusmc.nl	Tenured	Researcher	Applicant
Dr. J. Daemen	Erasmus MC	Cardiology	j.daemen@erasmusMc.nl	Tenured	Clinician	Applicant
Dr. ir. Theo (T.) van Walsum	Erasmus MC	Radiology/ Biomedical Imaging	t.vanwalsum@erasmusmc.nl	Tenured	Researcher	Applicant
Prof. dr. Kees (C.) Verhoef	Erasmus MC	Surgical Oncology	c.verhoef@erasmusmc.nl	Tenured	Clinician/ Researcher	Applicant
Dr. Stijn (S.) Keereweer	Erasmus MC	Otorhinolaryngology	s.keereweer@erasmusmc.nl	Tenured	Clinician/ Researcher	Applicant
Dr. Adriaan (A.) Moelker	Erasmus MC	Interventional Radiology	a.moelker@erasmusmc.nl	Tenured	Clinician/ Researcher	Applicant
Dr. Heleen (H.J.) van Beekhuizen	Erasmus MC	Gyneacological Oncology	h.vanbeekhuizen@erasmusmc.nl	Tenured	Clinician/ Researcher	Applicant

Dr. ir. Nick (N.J.) van de Berg,	Erasmus MC / TU Delft	Gyneacological Oncology Biomechanical Engineering	n.vandenberg@erasmusmc.nl		Researcher	Applicant
Prof. dr. Remi (R.A.) Nout	Erasmus MC	Radio therapy	r.nout@erasmusmc.nl	Tenured	Clinician/ Researcher	Applicant
Dr. Dalibor (D.) Vasilic	Erasmus MC	Plastic & Reconstruct. Surgery	d.vasilic@erasmusmc.nl	Tenured	Clinician/ Researcher	Applicant
Prof. dr. Clemens (C.M.F.) Dirven	Erasmus MC	Neurosurgery	c.dirven@erasmusmc.nl	Tenured	Clinician/ Researcher	Applicant
Dr. Sylvia Elkhuisen	Erasmus University ESHPM	Health Service Management and Organisation	elkhuisen@eshpm.eur.nl	Tenured		Applicant
Dr. Ken Redekop	Erasmus University, ESHPM	Health Technology Assessment	redekop@eshpm.eur.nl	Tenured		Applicant
Dr. Silvia Mamede	Erasmus MC/EUR, ESSB	Educational & Developmental Psychology	s.mamede@erasmusmc.nl	Tenured	Researcher	Applicant
Dr. Ir. Rik (H.J.) Vos	Erasmus MC / TU Delft	Biomedical Eng./ Medical Imaging	h.vos@erasmusmc.nl	Tenured	Researcher	Applicant
Prof. dr. ir. Paul (P.) Breedveld	TU Delft	Biomechanical Engineering	p.breedveld@tudelft.nl	Tenured		Applicant
Dr. Aimee Sakes	TU Delft	Biomechanical Engineering.	a.sakes@tudelft.nl	Tenure Track		Applicant
Dr. John (J.J.) van den Dobbelsteen,	TU Delft	Biomechanical Engineering	j.j.vandendobbelsteen@tudelft.nl	Tenured		Applicant
Prof. dr. Benno (B.H.W.) Hendriks	TU Delft / Philips	Biomechanical Engineering	b.h.w.hendriks@tudelft.nl	Tenured		Applicant
Dr. Frank (F.) Gijzen	Erasmus MC / TU Delft,	Biomed. Eng./ Biomechanical Eng.	f.gijzen@erasmusmc.nl	Tenured	Researcher	Applicant
Dr. Ricardo (R.) Marroquim	TU Delft	Comp. Graphics & visualisation	r.marroquim@tudelft.nl	Tenured		Applicant

Corresponding Lead: Prof. dr. Aad van der Lugt
 Planned start date of seed funding: 1-11-2022
 Primary Convergence theme: *Human Centered Technology and AI for Health*
 Secondary Convergence theme: *Improving Health Journeys*
 Keywords: *Minimally Invasive Interventions, Image Guidance, Augmented and Virtual Reality, Oncological and Cardiovascular Interventions, Brachytherapy, Minimally Invasive Surgery, Medical Device Development*

1.b Summary (299 words)

Objective: **Flagship Image-guided minimally invasive interventions (I-Guide)** will join forces to bring intraoperative imaging, sensing, steering and navigation technology into clinical practice enabling cardiologists, radiologists, radiation-oncologists, and surgeons to perform minimally invasive interventions with greater precision. These one-time-right treatments, without unnecessary trauma, will result in shorter hospital stay and reduce disease recurrence.

Design: Within **I-Guide** we will develop **novel devices and methods for (cardio-)vascular and oncological** clinical applications, focussing on:

1. preoperative and intraoperative imaging, e.g., ultrasound, virtual/augmented reality, endoscopy, and sensing (e.g., optical, impedance) for
2. guidance (e.g., user-friendly steering and navigation, image registration, instrument tracking and visualization) during
3. endovascular, minimally invasive surgical and radio-therapeutic interventions.

The **Technical Challenges** are developing innovative and affordable miniaturized devices and methods for image-guided navigation, sensing, and steering combined with intuitive interfaces.

Scientific impact: Combining our strong knowledge and experience in related research programmes will lead to a range of novel platform technologies in the fields of tissue sensing, steering, device-borne imaging, visualization/image fusion and navigation, leading to high-impact scientific publications, students educated, innovative concepts clinically evaluated, collaborative projects, new research grants, and spin-off companies.

Economic impact: The global minimally invasive surgical and cardiovascular instruments market is around USD30 billion/year and expected to grow by 50% the coming 6 to 7 years. The NL med-tech industry sector has a high R&D intensity and encompasses several companies (e.g., Philips Medical Systems) that are well equipped to turn new innovations into economic value. I-Guide also sparks the formation of new high-tech starters. There is a strong demand from medical end-users for new image guided treatment methods, based on novel devices. Together with the collaborating companies, I-Guide spans the complete innovation chain and can therefore react quickly to these new market demands, so that the Netherlands can profit from the related increased export/economic values.

1.c Public Summary (299 words)

Image-guided minimally invasive interventions result in more precise and safer procedures with less damage to healthy tissue and therefore quicker recovery of the patient. Image guidance supports the physician to precisely visualize and target the disease. These interventions are applied in an ever-expanding range of clinical scenarios, such as catheter and needle interventions and minimally invasive surgery. Devices are steered precisely to a target, deploying invasive and non-invasive sensors for assessment of local tissue properties. Advanced visualization tools merge image and sensory information, for instance as VR overlays on pre-operative images and planning.

Within the **Flagship Image-guided minimally invasive interventions (I-Guide)** we join forces to bring intraoperative imaging, sensing, steering and navigation technology into clinical practice. Cardiologists, radiologists, radiation-oncologists, and surgeons will be able to perform interventions more precisely, in full control. One-time-right treatments, sparing healthy tissue and function without unnecessary trauma, will result in shorter hospital stay and reduce disease recurrence.

I-Guide will focus on **cardiovascular, cerebrovascular, and oncological** clinical applications.

Novel devices and methods to be developed within **I-Guide** will focus on:

1. pre-operative and intraoperative imaging (e.g., US, VR/AR, endoscopy) and sensing (e.g., optical, impedance), for
2. guidance (e.g., user-friendly steering and navigation) of
3. endovascular, minimally invasive surgical and radio-therapeutic interventions.

The **Technical Challenges** of I-Guide are: the development of innovative and affordable medical devices integrating image-guided navigation, sensing, and steering at a miniature size. Intuitive interfaces will be easy to control by the surgeon or interventionist.

To bring the novel **I-Guide** devices to a **higher TRL (Technological Readiness Level)**, we will join experiences of the three involved institutes (TU Delft, Erasmus MC, EUR) on device prototyping, intraoperative imaging, clinical application, health technology assessments (HTA) and translation/valorisation. Moreover, together with small spin-offs, SMEs, and large companies, we will bring these innovative instruments into clinical practice.

2. Relevance to Convergence Health & Technology mission

(2500 words)

2.a Contribution to H&T mission and fit with H&T theme(s)

BACKGROUND AND FOCUS

I-Guide establishes an ecosystem for research into and clinical development of image-guided minimally invasive interventions. Minimally invasive medicine has radically transformed a range of clinical disciplines, enabling effective and safe treatments in a number of specific clinical applications, with the distinct advantages of fast recovery, minimal pain, trauma, and infection risk compared to open surgical approaches. High-quality image guidance is of critical importance for optimizing safety and efficacy of minimally invasive interventions. Lacking a direct visual link with the operator's working field, real-time, easy-to-use imaging, to visualize pathology, support device navigation, and characterize treatment effect, is key to inform decision making and tailor the intervention to the individual patient.

Technologies encompass visualization and navigation, image analysis, smart imaging and sensing instruments, and agile, thin minimally invasive devices. These broadly applicable technical concepts can be tailored to a range of clinical scenarios, of which we include four in this program, centred on vascular and oncological interventions. Economic, usability and learning aspects of the introduction of such technologies in a therapeutic setting need to be optimized to facilitate widespread adoption. The core group of 25 applicants at three knowledge institutes bring together the expertise to cover image-guided interventions from basic ideas to broad clinical deployment.

CONTRIBUTION TO THE H&T MISSION

I-Guide primarily contributes to the Convergence H&T mission **Individually tailored precise medical treatment for all**. Minimally invasive treatments offer great benefits to patients in the form of faster recovery, less trauma, less pain, and lower infection risk. The goal of the program is to make such interventions as versatile, safe, precise and predictable as possible. This includes personalization of interventional strategies, tailored to patient-specific anatomy and precise characterization of pathology for optimal outcomes, resulting in a decrease in disease recurrence.

For some interventions, devices may be customized, or patient-specific outcome predictions may inform the therapeutic approach. A continuous analysis of clinical needs will result in effective technological innovations and gradual and safe implementations in clinical practice.

The benefits to patients also translate into a potential for societal benefit. The economic benefits of faster recovery and shorter hospital stay, and less disease recurrences are evident. The translation to **Socioeconomic equity in health(care)** primarily offers challenges on a global scale. The specialized instruments that are needed are traditionally expensive and disposable. With a focus on reusability and accessibility, we introduce a relatively new concept in interventional device design. Within I-Guide cost-effective breakthroughs are anticipated through close collaboration between engineering, clinical and social scientists, from different disciplines of the three participating institutes.

FIT WITH THE H&T THEMES

Human-Centered Technology & AI for Health

This Flagship fits excellently within the **Operation Theatre 2030** and **Novel clinical imaging technology** foci of this theme. We will develop new interventions, interventional instruments, new imaging methods, and image analysis and navigation methods. The broad technical, clinical and socio/economic scope of the consortium stimulates crosspollination of different application spaces. Aligned with the philosophy of OT2030, integrated solutions and workflows will be designed, starting from end-user input.

Improving patient journeys

Image-guided interventions inherently combine part of the diagnostic trajectory with treatment. The operating physician can immediately act on the information provided by interventional imaging. Faster, more effective interventions with fewer complications benefit the patient and shorten the pathway back to health and full societal participation.

2.b Synergy between disciplines

The development of user-friendly and clinically effective I-Guide devices and tools involves a closely networked group of scientists and clinicians with a diversity of expertise and potential for truly transdisciplinary research. Project teams will combine problem-solving methods and skills from the medical, engineering, social and economic sciences. This unique breadth will lead to original, effective, systems-level solutions. We do not just introduce a new technology that subsequently may or may not be used, we consider the entire chain of technological possibility, economic viability, clinical adoption, and fit with the health care system.

All efforts within this program aim to realize at least clinical pilot research (TRL6) based on the innovations we investigate. Clinicians as end-users, engineers, and social scientists have been brought together to work closely together to jointly assess viable concepts, clinical needs, economic value and hurdles to implementation. Clinicians will describe the needs, test prototypes and assess the clinical value of technology. Scientists and engineers translate needs in the design process of devices and tools, build prototypes and interact with collaborative industrial partners.

Social scientists from the EUR will contribute by developing methodology to assess the value of technologies while the technologies are being developed. This allows for health technology assessments (HTAs) already performed in the early phase of instrument development, but also when new (pilot) technologies are implemented in health care practice. Merging approaches from medical device developments with insights from HTA will allow for steering the technology

development process in the right direction and continuously assess whether and how it is valuable to continue.

The 'Governance' section below describes the implementation of actions that will ensure mutual interactions and merging of activities within disciplines.

All initiatives in the flagship maintain close links with industry, both large-scale and small and medium enterprises (SMEs).

2.c Long term sustainability plan

Feasibility: I-Guide builds on the creation of the MedtechNL Device Lab, Phantom Lab, and (pre-)clinical testing facilities that will be built within the Erasmus MC-TU Delft Ecosystem. This infrastructure will ensure that the unique asset of I-Guide can be realized. I-Guide will spearhead image-guided interventions as one of the central themes of Convergence Health and Technology. It will integrate and expand an already vibrant research community on minimally invasive medical technologies. The joint research projects that are the subject of this application, as well as those that will be applied for in the near future, will strengthen those links. We consider the present team and research program as an effective open research collaboration. The funding in the present round will be deployed to initiate the research lines and establish a working consortium structure, which will be designed with expansion to new clinical applications and new engineering collaborators in mind. To facilitate this, we will reach out to the research communities within Convergence to broaden the scope of the program, based on new grants to be acquired, and to researchers and clinicians outside Convergence for collaboration and dissemination of our results. The aspects of (early-)HTA, usability and clinical decision-making research is relevant to all clinical/technical innovation programs incorporated herein, and the knowledge we build up here specifically for the field of minimally invasive interventions will attract collaborators.

The researchers that are involved from the start of the flagship program can build on an excellent international track record of the consortium in the field of technological and clinical innovations for minimally invasive applications. The funding requested for this flagship will be matched with a mix of publicly funded grants and private funding in the form of direct industry research contracts and co-financing. Young researchers appointed on the projects described in section 3.c will lay the groundwork for an integrated research program in which technologies find applications in different clinical fields and common knowledge is built up regarding the successful translation to clinical studies and beyond. We will develop the infrastructure and experience to perform such clinical trials with new instruments, developed in our institutes. Acquiring this capability is a demanding but necessary step in moving academic solutions for unmet needs into commercialization.

Risks within the first five years of central funding amount to:

- *Retaining talent, experience and knowledge:* postdocs and PhD students working on these technologies will acquire a unique and highly valuable set of skills and will be highly visible to and in high demand of (participating) industries.

Mitigation: The host institutions can accommodate e.g., tenure track positions in this field, in recognition of its versatility and importance, to secure their expertise. This year the Biomechanical Engineering group can attract a new tenure tracker, which is a good indicator for long term sustainability. In addition, we anticipate that some developments will be suitable for establishing spin-off companies, which may be led by former PhD students or postdocs of the flagship, augmenting the network with commercialization outlets.

- *Legal developments:* the medical device regulation, which governs the certification and use of medical devices in the EU, is a new law and there is limited jurisprudence on some of its grey areas, which are pertinent to the translational and (preclinical) research aspects of new technology. Particularly in the area of software, many things are still unclear. Conservative interpretations of the law may complicate the translational strategy of I-Guide.
Mitigation: We are in contact with the relevant governing bodies at the local and national levels to ensure that translational research remains viable.
- *Narrow base at EUR:* ESHPM and ESSB are new partners in this collaboration, integrating the social science perspective in the technology development and implementation process. EUR is currently only represented by four staff members with complementary expertise which may be vulnerable to personnel changes.
- *Mitigation:* From the start of the program we will engage more researchers and train PhD students. For long-term sustainability it is necessary that junior researchers can extend their employment post-PhD, beyond the first five years of central funding. Working together within this program strengthens the alignment of the two ESHPM departments and facilitate identification of funding opportunities.

Many participants in I-Guide are part of NIMIT, which is a >10 years collaboration within Medical Delta. There is a long-term collaboration between the Erasmus MC and TU Delft partners. They were, e.g., the leading partners working together within a large iMIT Perspectief program (2013-2019, 7.5MEuro). Moreover, members of the consortium have demonstrated to be able to attract external funding (see grant budget mentioned by the consortium members). This excellent track record in acquiring funding together with interest from companies forms a good basis for sustainability beyond the present funding period of 2022-2027. By that time, a number of young researchers will have been trained and will be candidates to expand the team or replace senior faculty that may leave.

2.d Visibility, attracting talents and new partners

Visibility. Visibility will be created through multiple ways. Staff, researchers, postdocs, PhDs, students will disseminate their work via:

- Publications in high visibility journals
- Sharing insights and experiences during presentations on national and international conferences academic conferences (iSMIT, CRAS, EAES, SPIE, IEEE, EuroPCR, ECR congresses, etc.)
- Multimedia (Facebook page, Twitter, YouTube, press releases, etc.).
- The I-Guide website that will be developed specially for this program.
- Interviews (e.g., local and national newspapers, NRC, Delta, Erasmus Magazine; professional periodicals).

Engaging BSc and MSc students. The large numbers of students supervised indicated in Pt. 4a (track record consortium members) show that the consortium is excellently connected to education programs. Most consortium members have already extensive experience in involving large numbers of students in their research. We will continue doing so within I-Guide. Hence within I-Guide:

- Students will conduct their BSc and MSc theses within I-GUIDE projects.
- Consortium members will share I-Guide experiences within educational programs. For example,
 - o Technology and Innovation within the ESHPM-Bachelor in Health Policy & Management,

- Device development for Biomedical Engineering and Biomechanical Design (Mechanical Engineering) students,
- Determining technical requirements for new devices and evaluation of devices in (pre) clinical practice for Clinical Technology, Technical Medicine, and Medicine students.

We experience that technology for minimally invasive applications is attractive for many students with different backgrounds. Within I-Guide these students from different disciplines will be motivated to collaborate together, e.g., during the research meetings and workshop days (see Section 4.d)

Attract Talents. To attract new talents, the vacant positions will be announced on international websites, such as academic transfer, communicated within our large international network. We also have good experience in engaging excellent (Cum Laude) MSc students, which allows us to identify talented PhD candidates.

Attract Partners. In the past years we already were able to attract a large number of partners in our eco-system, and regularly new partners connect to the consortium. Examples of connected partners are: Philips Medical System, Philips Research, Boston Scientific, Kaminari Medical BV, Synaptive, DEAM, DePuy, Dorc, Elekta, Oceanz, DSM, van Straten Medical, Surge-on Medical, Slam Ortho, MediShield, Layco Medic.

Attract Funding. The consortium has ample experience in attracting funding (see e.g., budget for grants mentioned in Section 4 (Consortium)). Some examples from the past are: iMIT Perspectief program (7.5 MEuro), VICI grants (GvS, PB), NWO-ZonMw-IMDI (Innovative Medical Devices Initiative) grants, TTW-OTP (Open Technology Program), and EU programs.

We also obtained a large number of valorisation grants: TTW-Demonstrator grants, TTW-Take-off (Phase 1&2) grants.

Also in the coming years, there are many possibilities for funding, such as OTP and TTW-Perspectief programs, current KIC-program on minimally invasive interventions, Groeifonds, EU programs, such as ITN, Horizon, Marie Curie, etc.

2.e Long term societal impact

The world-leading research groups involved in I-Guide represent a unique combination of clinical, technical and social expertise. To have *long term* scientific, economic and social impact, the existing ecosystem will be further strengthened and combined with researchers from Erasmus University. To bring the developed innovative **I-Guide** tools to a **higher TRL level**, we will further merge our experience on device prototyping, intraoperative imaging, clinical application, health technology assessments (HTA) and translation/valorisation of the three involved institutes. This together with the close collaboration with small spin-offs, MKB, and large companies will ensure that the developed solutions will be embedded in the health care setting to have long term impact.

Social impact: According to the WHO cardiovascular diseases are the leading cause of death globally, taking an estimated 17.9 million lives each year, ~32% of all deaths worldwide. Oncological diseases count for another 10 million lives [1]. Many of these patients can be cured if detected early and treated effectively. Because of these huge numbers, the impact of minimally invasive interventions for cardiovascular and oncological interventions is high. They typically permit shorter hospital stays, require less patient care and facilitate rapid postoperative recovery and therefore significantly fewer days of absence from work and more broadly, improved mobility, independence and better functioning in society [2].

Economic impact: The advantages of minimally invasive interventions over open techniques make them one of the most important developments in the healthcare industry. The global minimally invasive surgical instruments market is projected to reach USD 28.9 billion by 2026 from USD 20.2 billion in 2021, at a compound annual growth rate of 7.4% [3]. In 2020, the global interventional cardiology market size was valued at approximately \$11.1 billion and is expected to reach an astonishing \$16.2 billion in 2027 [4]. The Netherlands is one of the leaders in the development of new devices and procedures for minimally invasive interventions and the fast translation to the clinic. This Flagship supports Erasmus MC, EUR and TU Delft to continue playing a leading role in this challenging field.

Valorisation: The I-Guide pathway will cover the entire medical instrument development chain, from TRL1 to TRL 7-8 (demonstration phase), meaning that throughout the program there will be prototypes in different stages of the development. Our pathway will contribute continuously to new business opportunities for existing (large) Dutch and foreign companies by stimulating the industrial production value of medical technology in the Netherlands.

3. Flagship proposal (2000 words)

3.a Ambitions

The mission and ambition of I-Guide is to become world leading in the development of affordable innovative devices and guiding methods for minimally invasive procedures with an added value for patients, clinicians, science, society and business.

The **ambition** is to bring 5 devices/imaging systems at least 3 levels higher on TRLs, resulting in 5 new spin-off companies or new licences to existing companies, while maintaining high scientific output.

3.b Aims and feasibility

In the cardiovascular, cerebrovascular and oncological fields there is a strong need for innovation to further improve clinical outcomes following a minimally invasive approach. However, many technical solutions have difficulties finding their way into clinical practice. Convergence between Erasmus MC, EUR and TU Delft is essential for validation and efficacy research, especially in an early stage of validation, continued by (pre-)clinical validation, necessary for valorisation of the new image guided and minimally invasive technology.

The aim of the I-Guide Flagship is to accelerate the transition to image guided minimally invasive interventions in a close collaboration between clinicians, engineers and social scientists. I-Guide will develop innovative devices and systems for minimally invasive treatment of patients with **coronary artery disease, ischemic stroke and several cancers**, focussing on:

- **precision targeting with extremely slender catheters/needles that can steer and propel themselves** through delicate tissues and thin blood/lymphatic vessels,
- with highly **advanced imaging and sensing functionality at the tip** for instantaneous diagnosis of tissue properties,
- which provides sensing and imaging information necessary for intuitive navigation,

- and **advanced image processing methods** providing quantitative information for **therapeutic decision making**, and **navigation methods** using integrated pre-operative information and real-time imaging with **AR techniques** for visualization,
- that score excellent on health technology analyses (HTAs).

To support the latter aim, a new methodology will be developed to assess the value of technologies while the technologies are being developed. Such an early HTA will shed light on the expected direct (health outcomes, resource usage, etc.) and indirect consequences (operations, integrated planning, logistics, etc.) for different stakeholder groups.

Feasibility: I-Guide can build on an excellent (international) track record of the consortium in the field of technological and clinical innovations. Feasibility is supported by the intensive collaboration between parties with complementary expertise, and the eco systems with companies (see Section 4.c). The budget will partly be used to bring a number of devices to higher TRLs and to write new grant applications to expand our research.

Strategy to include students. For most consortium members it is already common to include BSc and MSc students in their research (see Section 4.a). They have access to students from medicine, technical medicine, clinical engineering, biomedical engineering, biomechanical design, healthcare management, and health economics. We will continue doing so and no specific strategy is needed to attract more students, because we already attract more students than we can handle. What could be improved is the collaboration between students. We will set-up 'JITs': joint interdisciplinary teams in which students from different disciplines will work together on a project.

3.c Work plan

Coherence of work plan:

Connections between the technical challenges and clinical applications are schematically shown in Figure 1.

Activities covered within I-Guide are indicated by the **arrow crossings** in Figure 2.

The three vertical technology pillars intersect with 5 clinical applications, with a key challenge at each crossing. The challenges that will be **focussed on in this call** (in PhD/postdoc projects) are indicated with a **Circle**. We envision that the developed technology will eventually be applied to other clinical applications as well. Devices will be developed, ready for clinical (pilot) trials in the MedtechNL DeviceLab (Groeifonds), to be established at the Convergence centres in 2023. The EUR group will assess the value of technologies during their development (yellow area, Figure 2). A shared verification and validation infrastructure will be equipped to test instruments performance (blue area, Figure 2).

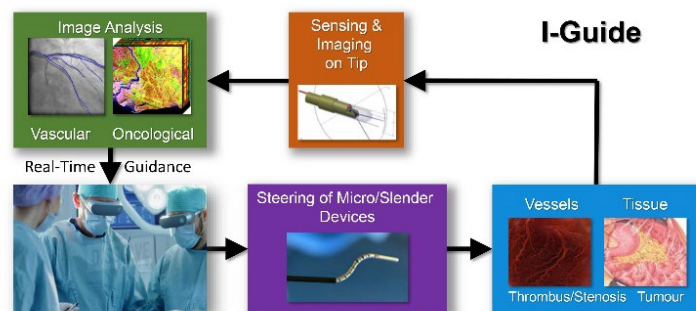


Figure 1. Connection of main constituents of I-Guide.
Technical challenges: purple, orange, green, and
clinical applications: blue.

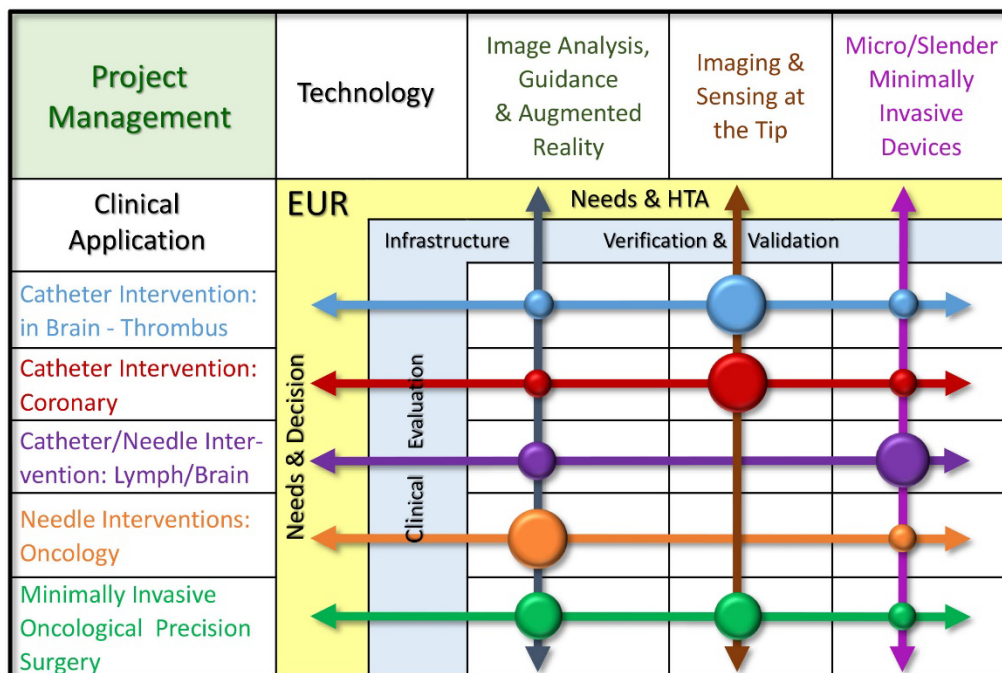


Figure 2. The I-Guide WP scheme. Colored circles represent the WPs focal points and their size the level of involvement with the technical pillars.

Imaging and sensing for catheter intervention in the brain (Blue Circles) (TRL3 -> TRL6)

Background: Affordable, disposable sensing devices will be developed for intraprocedural tissue characterization in the brain with a (steerable) guidewire or microcatheter.

Objective: Optical (DRS) characterization of tumours and thrombus; impedance spectroscopy for bioelectricity.

Long-term work plan & deliverables:

- Thrombus biomarkers: support therapeutic strategy choice with optical/mechanical thrombus characterization, integrated in a guidewire or microcatheter.
- Impedance spectroscopy applied to ablation lesion depth characterization.
- Design for manufacturability of interventional sensing devices.

Precision Percutaneous Coronary Interventions (Red Circles) (TRL4 -> TRL7)

Background: In NL alone, yearly, almost 50000 percutaneous coronary interventions (PCI) are performed. Despite proven patient benefits, only 5% of PCIs are guided by intravascular imaging. The Cardiology Department of Erasmus MC has accumulated a database containing >20000 image-data sets.

Objective: Developing technologies and workflows to simplify the use of intravascular (IV)-imaging.

Long-term work plan & deliverables (capitalizing on the new Erasmus MC IV-imaging core lab):

- Physics-based and data-driven image analysis methods for patient-specific predictive modelling, pre-empting adverse outcomes. Decision-making heuristics based on these new analyses will be tested in prospective studies.
- Direct quantitative visualization of coronary artery flow patterns for interventional use will be developed based on IV ultrasound and IV-OCT.
- We will study the decision-making process based on these images in a real-world interventional setting, build-up and transfer expertise (with EUR).

Steering Ultrathin Minimally Invasive Devices (Purple Circles) (TRL2 ->TRL5)

Background: Metastasis is the primary cause of cancer-related deaths. Most epithelial cancers first develop metastatic growth by spreading via lymphatic vessels and nodes. Surgical excision of a sentinel node or entire lymph node package removes lymphatic tissue without prior knowledge of its metastatic condition.

Objective: Preventing (diagnostic) excision of otherwise healthy lymph nodes, we aim to develop ultrathin (<0.3 mm) catheters taking multiple micro-biopsies from within the lymph nodes without distorting local lymphatic anatomy. The catheters will be securely navigated towards designated lymph nodes utilizing merged image information obtained prior to (ultrasound, CTA, MRA) and during (NIRF) the biopsy. A robotically-controlled digital microscope offers improved 3D visualization.

Long-term work plan & deliverables:

- Deliver an ultrathin catheter self-propelling micro-biopsy catheter
- Integrate a signal emitting probe to guide the tip of the catheter to the designated lymph node.

Precision Ablation of Liver Tumours (Orange Circles, 1) (TRL4 ->TRL7):

Background: CT-guided ablation of liver tumours is a minimally invasive therapy for liver cancer patients not eligible for liver surgery. The physician introduces a needle into the tumour, subsequently destroying it by local heating using RF or MW radiation. Positioning the needle percutaneously into the tumour remains a challenge due to limitations of pre-operative and interventional imaging, and the difficulty of mentally mapping information derived from these imaging modalities.

Objective: to develop and assess a novel technique for CT-image guidance for percutaneous ablation of liver tumours.

Long term work plan & deliverables:

- Assessment of a previously developed software-based approach for improved image guidance in a randomized clinical trial. Patients will be treated either conventionally or with the new approach; the main outcome being (local) tumour control during follow-up.
- Quantitative measures developed to assess the ablation result (by combining pre- and intra-operative imaging) and use for therapeutic decision making and to estimate the risk of tumour recurrence after ablation.

Real-Time Image-Guided Adaptive Brachytherapy in Pelvic Cancers (Orange Circles, 2) (TRL4->TRL6):

Background: Minimally invasive devices (needles, catheters) are used for localized placement of radioactive seeds. Recently, 3D image-guidance and precise radiation dose adaptations have led to a dramatic increase in tumour control with less side effects for gynecological and urological cancer patients.

Objective: to optimize the workflow and employment of image-guided tools in brachytherapy, while keeping health care sustainable and making procedures more labor friendly.

Long term work plan & Deliverables:

- Integrating HTA by assessment of image-guided workflows along the full patient journey (pre-planning, implantation, treatment).
- Improving and aligning automatized steps in needle path-planning, reconstruction, segmentation, dose planning and QA of delivery, and integrating these in a time and user-friendly workflow.
- Improving implant quality by needle steering in target tissue and optimizing image-guidance.

Minimally Invasive Oncological Precision Surgery (Green Dots) (TRL4 ->TRL6-7):

Background: In cancer surgery, a clear surgical margin is the main prognostic factor for survival. However, current intraoperative tools are often insufficient for identifying and completely removing tumours, leading to under-treatment (incomplete tumour removal), and overtreatment (removing too much healthy tissue to obtain clear margins).

Objective: Realizing minimally invasive precision surgery to improve clinical, functional, and oncological outcomes of patients with head/neck cancer or ovarian cancer (target applications), by combining whole-field imaging (hyperspectral, fluorescence) and at-the-tip sensing (Raman, DRS, impedance).

Long term work plan & deliverables:

- Develop, optimize and assess the effectiveness of tissue classification methods based on these intraoperative imaging and sensing tools.
- Evaluate dedicated (flexible) minimally invasive devices like probes, needles, endoscopes or laparoscopes with embedded optical fibres
- Visualize tissue classifications with augmented reality.

Validation Translation (Blue Area):

Background: Validation is key in clinical translation of the technologies developed in the vertical pillars. Pre-clinical in-vitro validations form an essential and effective way to optimize device design. Appropriate patient material, relevant geometrical models and in-vitro evaluation with (pre-)clinical imaging facilities greatly enhance translation. Moreover, evaluations in increasingly realistic environments are needed to ascend the TRL ladder.

Objective: To develop a new platform, building on current facilities at the Erasmus MC including imaging facilities, mechanical testing equipment and quantitative histology. We plan to mirror basic facilities to TU Delft, enriching the existing labs with dedicated high-end equipment reflecting their expertise.

Long term work plan/Deliverables: Invest in securing personal technical and clinical/biological expertise. The joint facilities and expertise will enable us to exchange students and researchers between the institutes, providing them with hardware support and technical knowledge.

Early Health Technology Assessment (Yellow Area):

Background: Technology development in healthcare involves various stakeholder groups and a multitude of (sequential) decisions. Traditionally, HTA is done when new (pilot) technologies are implemented in health care practice. Our ambition is to develop methodology to assess the value of technologies during their development. This will shed light on the expected direct (health outcomes, resource usage, etc.) and indirect consequences (operations, integrated planning, logistics, etc.) for different stakeholder groups and help to improve decision-making by answering the question "Is it valuable to continue and in which direction?".

Objective: Develop methodology to assess the value of technologies during their development.

Long term work plan & Deliverables: A methodology will be developed based on inputs from use cases (e.g. brachytherapy, percutaneous coronary interventions). This will allow us: 1. to identify stakeholders and assess the needs, 2. to assess feasibility of early data provision, and 3. to explore how to communicate and present the data in an informative way for decision-making.

3.d Alignment of the program with relevant national and international research agendas

I-Guide fits well within (inter)national research agendas, indicated by:

- **NWA:** How can we develop minimally invasive techniques and interventions for diagnosis, prognosis and treatment. New devices are needed for safe removal of tumours, and early treatment of chronic diseases (e.g., early cartilage treatment in joints), cardiovascular disease ; all part of personalized medicine track.
- **NFU:** “Om naast fundamentele wetenschap ...zijn ondersteunende technologieën relevant: en (minimaal-invasief) toedieningsinstrumentarium”; “Hoe ontwikkelen we minimaal-invasieve technieken en interventies voor de diagnose, prognose en behandeling van patiënten?”; “Hoe kunnen we de meest geavanceerde microscopietechnieken incorporeren in miniatuurgereedschappen voor minimaal-invasieve diagnose en behandeling van ziekten?”
- **Topsector:** The proposed innovations in minimally invasive technologies are also explicitly demanded within the Health care roadmaps of top sectors High Tech Systems and Materials (HTSM) and Image guided interventions of LifeSciences & Health (LS&H).
- **Groeifonds MedtechNL:** Several core members of the I-Guide team are leading parts of the MedtechNL Groeifonds consortium. Activities in the intramural care segment of that program closely parallel and vitally support the efforts described here.
- **International:** I-Guide is aligned with EU (e.g. Marie Curie ITNs ATLAS, AuTonomous intraLuminAI Surgeryand Intraluminal navigation <https://atlas-itn.eu/>; MIREIA: Mixed Reality in medical Education based on Interactive Applications, <http://www.mireia-project.eu/>. Projects in I-Guide will contribute to EU’s Cancer mission and several programs within Horizon Europe.

4. Consortium (2000 words)

4.a Track record of the consortium members

For each consortium member, two key publications are listed in Annex 4.

Prof. dr. A. van der Lugt: 15 years of experience in the evaluation of endovascular treatment of neurovascular disease. PI of the MR CLEAN study which had an impact on acute stroke treatment worldwide with a change in guidelines. Grants obtained 7.5MEuro, 28 PhDs graduated, and currently supervising 10 PhD, >500 scientific publications (6x NEJM, 2x Lancet). Past/present collaborative projects with Stryker, Medtronic, Penumbra, Cerenovus. Research leader CONTRAST Consortium 2022-2027.

Prof. dr. J. Dankelman: >25 years of research experience in Minimally Invasive Surgery and Interventional techniques, Grants obtained ~15MEuro, Supervision: >50 BSc, >200 MSc, 36 PhDs, currently supervising 16 PhD, (co)-published>220 scientific publications. Steering committee member of SMIT. Lead iMIT Perspectief program. Member of Royal Dutch Academy of Sciences. Spinoff companies: MediShield (Smooth Seton/ Force Sense system), Surge-on Medical (Steerable punch), Layco (video laryngoscope), Mint Zorg-advies, GreenCycl, Simendo.

Dr. Sandra Sülz: >10 years of research and teaching experience in health service operations management, award-winning publication (Eugen-Münch Award for innovative health service research), supervision of >25 MSc students.

Prof. dr. Gijs van Soest: 15 years of experience in intravascular/minimally invasive imaging with optics and ultrasound: ± 8.5 M€ in grants obtained. Supervision of 15 PhDs, 20 BSc and 10 MSc students. Founder and scientific advisor of Kaminari Medical BV (intravascular photoacoustics), past/present collaborative projects with Boston Scientific, Philips, Abbott, Oldelft, Visualsonics, Mindray, etc. Conference chair of SPIE Diagnostic and Therapeutic Applications of Light in Cardiology, founder/scientific director of Optics in Cardiology, from 2022 Image guidance in PCI (iPCI). Member of the Scientific Board of the Netherlands Heart Institute.

Prof. dr. A.F.W. van der Steen: >30 years of experience in catheter-based technologies. Supervision of > 60 PhD theses. Head of Biomedical Engineering of the Thorax centre ErasmusMC Rotterdam; professor in Physics at the TUDelft; Honorary visiting professor to Chinese Academy of Sciences. 2000 NWO PIONIER technical sciences and 2007 Simon Stevin Master. Fellow of IEEE and the European Society of Cardiology. Royal Dutch Academy of Sciences board member and Netherlands Academy of Technology and Innovation member. He has worked his whole career at the crossroads between physics, technology and health. Cofounder and past president of the Medical Delta.

Dr. J. Daemen, interventional cardiologist, PhD Cum Laude, participates in cath lab related activities including complex PCI and use of left ventricular assist devices, intracoronary imaging, and structural heart interventions. Specific focus on intracoronary imaging and physiology and renal sympathetic denervation. Co-authored >250 publications and is a fellow of the European Society of Cardiology (FESC)

Dr. ir. Theo van Walsum: 20 years expertise in image analysis for therapeutic decision making, image-based navigation, tracking patient motion and deformation, and integrating pre-operative information in minimally invasive interventions. (co-)published > 100 scientific articles, several patents. Supervised >30 BSc, >25 MSc, >12 PhDs, currently supervising 6 PhDs. Obtained 1.1 MEuro funding as PI (TKI projects), 2 MEuro funding as co-PI (H2020, STW demonstrator). Collaborating with PIE MI, Quantib and Philips Healthcare. Active in CONTRAST, a national consortium for stroke research.

Prof. dr. C. Verhoef: Expertise: over 20 years in surgical oncology and clinical and translational research in patients with advanced sarcoma, melanoma, advanced colorectal cancer and breast cancer. Common denominators are new treatment modalities and determination of prognostic factors. Board member of several international Oncology & Research associations. Grants obtained ~ 6 MEuro; supervision: PhDs > 20 graduated; supervision > 30

Dr. Stijn Keereweer is a full-time Head and Neck Oncologic Surgeon focusing on development and use of image-guided surgery using fluorescent tumor-specific targeting agents and other spectroscopic techniques in preclinical and clinical trials. This includes the quest for new targeting agents, optimization of intraoperative camera systems, and clinical translation. ~ 1 M€ as PI and has been granted 4 personal awards. Currently supervising 3 PhDs and a PostDoc.

Dr. Adriaan Moelker is expert in minimally invasive local treatment of both oncologic and non-oncologic patients in terms of image guidance and treatment outcome. (Co-)published 200 scientific articles. Co-promotor of 8 PhDs. Involved in funding acquisitions for several scientific projects including industry sponsored (education grants) studies (Cook Medical, Getinge, QMedics).

Dr H.J. van Beekhuizen: MD, PhD, MSc clin trials, gynaecological oncologist. 15 years of experience in tertiary gyn oncological surgery and studies. Grants obtained 1.8M. Supervision 15 MSc, 6 PhDs students.

Dr. ir. N.J. van de Berg: Expertise: 10 years in developing image-guided tools for oncological surgery and interventions. Grants obtained ~900kEuro, supervision: >30 BSc/MSc, 2 PhDs.

Dr. D. Vasilic: >10 years of experience in complex microsurgical reconstructions, dedicated to pathology of lymphatic system, pioneered the (supra)microsurgical treatments of lymphedema, spearheads conversion to digital robotic microscopy. Grants obtained 1,2 M€. Supervision >15 MSc, 5 PhDs, 1 PostDoc

Prof. dr. Clemens Dirven: >20 years of expertise in neurosurgical oncology, co-founded the Dutch brain tumor center at ErasmusMC. Common denominator: translational surgical neuro-oncology with a focus on novel therapies, local delivery, and improvement of neurosurgical and microsurgical results.

Dr. Sylvia Elkhuisen: >13 years of research and teaching experience in health service operations management, supervision of >80 MSc, 1 PhDs.

Dr. Ken Redekop: >20 years of experience in observational research, clinical trial analysis, and medical technology assessment. >60 MSc/BSc Students, 13PhDs, EU Horizon SPCCT and Optogenerapy (~6MEuro per project, 300kEuro for EUR).

Dr. Sílvia Mamede: 15 years of experience in research on clinical reasoning, cognitive diagnostic errors, and clinical reasoning education; (co-)supervision 14 PhDs (7 completed; 7 ongoing).

Dr. H.J. Vos: MSc Applied Physics Cum Laude, associate professor ErasmusMC and TUDelft for high frame rate echography. Received Dutch NWO-TTW-VIDI grant. Research interests: acoustical array technology for biomedical imaging in all its aspects: transducers, 2-D and 3-D beamforming, cardiac shear waves, ultrafast Doppler, contrast imaging, and related subclinical and clinical studies. Co-supervision 10 PhDs (6 completed, 4 ongoing), currently PI for four projects, 500k€–1M€ each.

Prof. Dr. ir. Paul Breedveld (MSc Cum Laude, PhD Cum Laude, AvL distinguished professor, >100 MSc, 20 PhDs). Co-chair 4TU Soft Robotics. Received many awards, KNAW and VICI grant, involved in numerous EU and NWO projects with companies such as Philips and DORC, his research focuses on the application of smart biological concepts in innovative surgical devices which led to spin-offs Simendo (laparoscopic trainers) and DEAM (squid-inspired steerable instruments).

Dr. ir. Aimée Sakes: 10-year experience in development of bio-inspired soft instrumentation for minimally invasive surgery. Developed a range of novel prototypes, among which a series of innovative impulse catheters, the world's first steerable 3D-printed bipolar electrosurgical grasper, and a new type of transport mechanism inspired by the ovipositor (egg-laying needle) of parasitic wasps. Supervised >20 BSc, >30 MSc, currently 5 PhDs.

Dr. John J. Van den Dobbelsteen >20 years of research experience, co-authored >130 journal papers, secured >11MEuro on external funding, director of studies Technical Medicine, Vice-chair human research ethical committee TU Delft, (co)promotor of 12 PhDs, current supervised 6 PhDs and 3 fte support staff. Steering committee member of iSMIT.

Dr. Frank Gijzen: 20 years of experience in cardiovascular imaging and biomechanics, works at EMC (0.5fte) and TUD (0.5fte). Supervised > 50 BSc/MSc and 7 PhDs, secured ~3M€, involved in large EU project focused at increasing TRL levels of newly developed in silico products for clinical applications. Collaborates with Philips, Pie Medical, MiLabs, Boston Scientific, working on integrating biomechanical modelling and device design with advanced imaging.

Prof. dr. ir. Benno Hendriks: > 30 years of experience working at Philips and 6 years (1 day/wk) at TUDelft, developed surgical navigations system -> product Philips ClarifEye, developed optical sensing for tissue discrimination at tip of instruments for oncology and spine surgery. Obtained as Co-PI ~3MEuro funding. Supervising/supervised 7 PhDs and >20 MSc.

Dr. Ricardo Marroquim: broad interest in visual computing topics but specifically in methods to capture and digitally reproduce the real world's appearance and interactivity and visualization and object tracking in mixed-reality. Involved in many projects involving AR/VR, such as for sports, education and medical procedures.

Attract funding beyond 2026: The collective funding track record of the consortium, combined with the excellent young new researchers and new opportunities offered by the Flagship collaboration, ensures that the consortium will attract ample financing also after the first 5 years.

4.b Expertise of the consortium members in relation to the aims and objectives of the program

The track records of the consortium members (Section 4a) show that there is an excellent mix of clinical (surgical, cardiovascular, neurological), engineering (device developers, imaging scientists, human systems interaction) and social scientists (health economics, psychology) with relevant expertise participating in I-Guide. Moreover, there is a lot of diversity in experience within the consortium. It has members with >25 years of experience working in the engineering and clinical field of minimally invasive interventions. The consortium also attracted enthusiastic researchers from the social sciences, having experience in e.g., operations management and medical technology assessment.

4.c Involvement of partners

The I-Guide members have long term collaborations with several small, medium and large companies connected to their research. There is close collaboration with

- Large/medium companies, such as Philips Research (via the part-time position of Prof. Benno Hendriks at TU Delft and research fellow at Philips), Philips Medical System, Elekta, Dorc, Demcon, Boston Scientific, van Straten Medical, DePuys.
- Spin-of companies, such as Kaminari Medical, DEAM, Surge-on Medical, MediShield, Layco Medical.

Many of these companies already participate in current projects or have indicated willingness to participate in future projects. The stage of involvement varies from early stage prototype (e.g. with Philips), to pre-clinical evaluation, (e.g. Layco Medical), to clinical evaluation (e.g. Synaptive). The MedtechNL Transition Centre for Minimally-invasive Interventions, which includes the DeviceLab, fulfils a key role for much of the research in this flagship. Ten companies have pledged their support for this new innovation infrastructure.

4.d Governance, organization and collaboration strategy of the consortium

I-Guide is organised according to the technical pillars and clinical horizontal rows indicated in the figure of Section 3.c. To build up a closely-knit research community around I-Guide we will need frequent interactions at all levels, resulting in exchange of information/knowledge and mutual assistance. Moreover, we will set-up a research environment where knowledge and facilities of the different institutes are collectively and, therefore, more efficiently used.

- The *I-GUIDE Board (IB)* will be responsible for the successful implementation of collaborative research and valorisation activities in the consortium. IB will consist of the leads of the three participating institutes, and the SB, CB, and SAB (see below).
- The *Scientific Board (SB)* will be formed by the leaders of the pillar WPs and the horizontal WPs. The SB activities will be coordinated through formal meetings of the SB every 3 months. The SB meetings will discuss: 1. Project progress, milestones and deliverables; risk monitoring, 2. Scientific and administrative matters arising, 3. Strategies for dissemination, publications and communications, 4. IP regulations and licensing, 5. Valorisation strategy and 6. Opportunities for new research/collaborations/grants.
- The *Clinical Board (CB)* will review and make recommendations on clinical, regulatory and ethical aspects of the project. It will comprise two clinical researchers from I-Guide, one clinician from another centre, and a medical device regulatory expert.
- The *Social Advisory Board (SAB)* will advise on ethical training and social issues that may arise during the project. It will include representation of patients, nurses and allied professionals, insurers and healthcare governing bodies.
- The *Advisory Board (IAB)* will be a panel of 4 external experts from different disciplines (e.g., from SME) supporting the programme. The IAB will participate in annual meetings and advise the IB on aspects as technologies, including the utilization plan, workflows and new clinical procedures developed. The IAB will meet twice a year

A closely collaborating network will be facilitated by the following activities on the program level:

- Each year an I-Guide research meeting/workshop day will be organized for the PhDs, PostDocs and other researchers involved in I-Guide. Each PhD/PostDoc will present their work with time for discussion. One or two renowned (internal) scientists will be invited as well.
- Halfway through the project, in the middle of the 3th year, a midterm symposium will be organised with workshops, presentations and poster sessions, on scientific and on valorisation subjects.
- Every 2 years a conference will be organised to summarize the overall outcomes and discuss (commercial) implementation towards the future.

Annexes (do not count toward maximum number of pages)

I Budget table (including the external funding table)

A pluriannual plan will be required if your proposal is accepted

Overview of External Funds:

Current Grants

- VICI Van Soest. Funder: NWO. Fit with I-Guide: 33% (intravascular), 2018 – 2024.
Companies: **Mindray Zonare, Terumo, FUJIFILM VisualSonics, Waters,**
Budget: 1.9 M€ of which 170 k€ in-kind cofounding.
- PICA-Heart, Funder: LSH-TKI. Fit with I-Guide: 25% (intravascular), 2021 – 2024.
Companies: **Percuro BV, PanaceAR BV, FUJIFILM Visualsonics and Kaminari Medical BV.**
Budget: 809 k€ of which 78 k€ in-kind and 70 k€ in-cash cofounding.
- Automated quantification of calcium in IVUS. Fit with I-Guide: 100%, 2022 – 2024
Funder: **Boston Scientific** (direct industrial grant).
Budget: 220 k€ private cash-fundin.
- NWO OTP: Steering through the Vertebrae. Fit with I-Guide 50%, 2020 - 2024,
Companies: **Philips Research, DEAM.**
Budget: 752 k€ of which 102 k€ in-kind and 20 k€ in-cash cofounding.
- MEDPHOT: NWO Perspectief, 2021 - 2025.
Companies: **Elesta, Percuro,**
Budget: 494k€ of which 42k€ in-kind and 92k€ in-cash cofounding.
- NWO-TTW-KWF Architect, Fit with I-Guide 100%, 2021 – 2025.
Companies: **Elekta, Oceans, DSM Oceanz**
Budget: 703K€ of which 64k€ in-kind and 25k€ in-cash cofounding.
- NWO AES VIDI (dr. van der Heiden), 2021 - 2026
Budget: 1003k€.
- TKI-LSH DIMIROS Lymph vessel surgery. 2021 - 2025,
Company: **Synaptive**
Budget: 823k€ with 335k€ in-kind cofounding.
- ZonMw IMDI Sensing in Surgery, extended till end-2023
Company: **Philips research,**
Budget: 320k€ with 220k€ in-kind cofounding
- TKI-LSH Q-Maestro, Fit with I-Guide: 100%, 2020-2024.
Companies: **PPP, Philips Healthcare,**
Budget: 1 M€, with 200 k€ in-cash and 100 k€ in-kind cofounding.
- TKI-LSH 700 kEuro, Fit with I-Guide: 100%, till mid-2024,
Company: **Philips**
Budget: 500 k€ with 200 k€ in-cash and 100 k€ in-kind cofounding.
- ICAI Stroke Lab, NWO, 5 year, Fit in I-Guide 40%
Company: **Philips**
Budget: 880 k€ with 200 k€ in-kind and 800 k€ in-cash cofounding.

Grants applications submitted:

- Next-gen MI-OCT, Funder: LSH-TKI, Companies: **Kinetron BV, IDE group BV, MPT Europe BV**, Budget: 524 k€ of which 89 k€ private in-kind cofounding, Fit with I-Guide: 100%, 2022-2026.
- MedTechNL: Device Lab, Funder: Nationale Groeifonds, Fit with I-Guide: 100%. Companies: **Philips, Demcon, PIE Medical, TNO**, 2022-2030; carrier project 2024-2027, Budget: 4.3 M€; additional private cofounding to projects running in this facility.
- ICAI Stroke Lab as part of Robust LTP. Funder: NWO, Company: **Philips Healthcare**, Budget: 2.2 M€, 800 k€ in-cash, 200 k€ in-kind, Fit with I-Guide: 40%. 2023-2027
- I-Guide: Budget: **Philips Research**: in-kind 100k€, in-cash, 80k, **DEAM**: in-kind 60k€

Future Grant application (planned)

- NWO VENI: A.Sakes. Budget: €350k€. Intended project period: 2023-2027
- EIC Pathfinder Open: Budget: €3M€. **Rev1Engineering**, period: 2023-2028
- NWO OTP Inside-Out: Companies: **BMEP, DEAM, HollandPTC**. 2023-2027 Budget: €700k, Matching €45k€.
- ZonMW Open Technology, Companies: TBD, Fit with I-Guide: 50%, 2023-2028 Budget: 800 k€ of which 150 k€ in-kind private cofounding,
- KIC Key Enabling Technologies for Minimally Invasive Interventions in Healthcare: OCT, Funder: NWO/KIC. Companies: TBD (**Abbott, Philips, startups**), Budget: 1.1 M€ of which 30% cofounding, Fit with I-Guide: 100%, 2023-2028
- KIC Key Enabling Technologies for Minimally Invasive Interventions in Healthcare: metamaterial devices. Funder: NWO/KIC. Companies: TBD. 2023-2027 Budget: 850 k€ of which 30% cofounding. Fit with I-Guide: 100%
- EIC Accelerator Open. Funder: EIC, Company: **Kaminari Medical** (beneficiary), Budget: 2.5 M€. Fit I-Guide: 50%, 2022-2024
- Hartstichting/ZonMW: Image-based decision making in MI interventions. Budget: TBD
- KWF grant, Oncological surgery, 500k€, 2023-2027

II Letters of intent and letters of commitment

1. Letter of Support/Intent AngioCare
2. Letter of Support/Intent HS Hospital Services S.p.A.
3. Letter of Support/Intent Pie Medical Imaging
4. Letter of Intent DEAM
5. Letter of Intent Synaptiv
6. Letter of Intent Kaminari Medical
7. Letter of Intent Philips

In Annex I above, the **companies involved** in current projects are indicated in **bold**.

Commitment letters of these **current projects** have not been included because of confidentiality issues, but can be consulted, on request.

III Letter of support from department head of the main lead

Letter from department head is attached.

IV References

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